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MelodyMemo: An Emotion-Aware Memory–Music Modeling and Recommendation System for Older Adults

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Abstract

Music strongly evokes autobiographical memories and emotional responses, yet existing recommendation systems do not preserve emotional context from personal narratives. This project presents MelodyMemo, an emotion-aware framework integrating natural language processing with hybrid recommender systems for emotionally aligned music selection for older adults. Two hypotheses were tested: (H1) a pretrained transformer classifier achieves $\geq 75\%$ emotion classification accuracy, and (H2) a hybrid recommendation algorithm improves emotional alignment compared with individual methods.

The system consists of two modules. R1 (Emotion Extraction) applies a pretrained DistilRoBERTa classifier to analyze autobiographical stories, infer emotional categories, and map them to musical attributes (tempo, valence, energy). R2 (Music Recommendation) takes the derived attributes and generates song recommendations using collaborative filtering, content-based filtering, and a hybrid approach (50% collaborative, 50% content-based). Evaluation used 50 emotion-targeted autobiographical narratives, a curated catalog of 50 era-appropriate songs, and simulated profiles for 25 older adult users.

R1 achieved 80% classification accuracy across seven emotion categories, exceeding the 75% threshold (H1 confirmed). Cross-model comparisons using GPT-4o and Gemini-2.5-Flash indicated consistent challenges for subtle categories such as love and neutral. The hybrid recommender achieved the highest average recommendation score (0.941), outperforming content-based filtering, collaborative filtering, and a random baseline (0.503), confirming H2. Paired per-user comparisons confirmed statistical significance ($p < 0.001$). An ablation study confirmed robustness across weighting parameters.

These findings demonstrate that integrating emotion extraction with hybrid recommendation can preserve emotional alignment between autobiographical memory and music, providing a foundation for future music-based interventions for older adults.

1. Introduction

1.1 Background

Older adults represent one of the fastest-growing demographic groups in the United States, with the population aged 65 and older increasing rapidly in recent decades (U.S. Census Bureau, 2020). At the same time, social isolation and loneliness have emerged as major public health concerns, with substantial proportions of older adults reporting reduced social and emotional connection (CDC, n.d.; National Academies of Sciences, 2020). These challenges motivate the exploration of technology-supported music-based interventions designed to promote emotional engagement and autobiographical memory recall.

Music has long been recognized as a powerful stimulus for memory retrieval. Janata et al. (2007) characterized music-evoked autobiographical memories (MEAMs) — vivid personal memories triggered by familiar music — and demonstrated their strong emotional associations. Subsequent research by Cunningham et al. (2019) confirmed that MEAMs remain robust in older adults, including those with early-stage cognitive impairment, suggesting that music-based interventions could serve as therapeutic tools for this population.

Despite these findings, to our knowledge, no prior system has systematically combined autobiographical narrative text with emotionally matched music recommendations. Current music platforms such as Spotify optimize for preference similarity — recommending songs based on listening history and collaborative filtering among users with similar tastes. This approach does not account for the emotional content of personal memories or the therapeutic goal of preserving emotional alignment between a person’s story and the music they hear.

1.2 Research Gap

The intersection of three research areas — natural language processing (NLP), music information retrieval (MIR), and music-based interventions motivated by music therapy research — presents an unexplored opportunity. Transformer-based language models have demonstrated strong performance in emotion classification tasks (Hartmann, 2022), while recommender systems have been widely developed for domains such as movies and consumer products (Koren et al., 2009). Despite these advances, prior work has largely treated emotion recognition and music recommendation as independent problems. To our knowledge, no existing study integrates narrative emotion extraction with music recommendation to form an emotion-preserving pipeline tailored to autobiographical memory contexts.

1.3 Research Questions

This study addresses two primary research questions:

1. Can a pre-trained transformer model reliably extract emotions from autobiographical narrative text?
2. Can a hybrid recommendation algorithm outperform individual algorithms and a random baseline in preserving emotional alignment between stories and recommended songs?

1.4 Hypotheses

- **H1:** A DistilRoBERTa transformer model can classify emotions in autobiographical narrative text with $\geq 75\%$ accuracy across seven emotion categories. The 75% threshold was chosen as a practical performance target because the pretrained emotion model was evaluated on autobiographical narratives that differ substantially from the datasets used during training.
- **H2:** A hybrid recommendation algorithm combining collaborative filtering and content-based filtering outperforms each individual algorithm and a random baseline on emotion-match rate and recommendation score.

1.5 Novel Contributions

This study makes three novel contributions:

1. **Narrative emotion extraction** — applying transformer-based NLP to autobiographical memory text, a domain distinct from the social media and product review corpora on which these models are typically evaluated.
 2. **Emotion-to-musical-feature mapping** — creating a systematic bridge between detected narrative emotions and musical attributes (tempo, valence, energy), connecting psychology and music information retrieval.
 3. **Hybrid recommendation optimization** — combining collaborative and content-based filtering with an explicit goal of emotion preservation rather than preference prediction.
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2. Materials and Methods

2.1 System Architecture

MelodyMemo is an experimental research platform consisting of two modules arranged in a sequential pipeline (Figure 1):

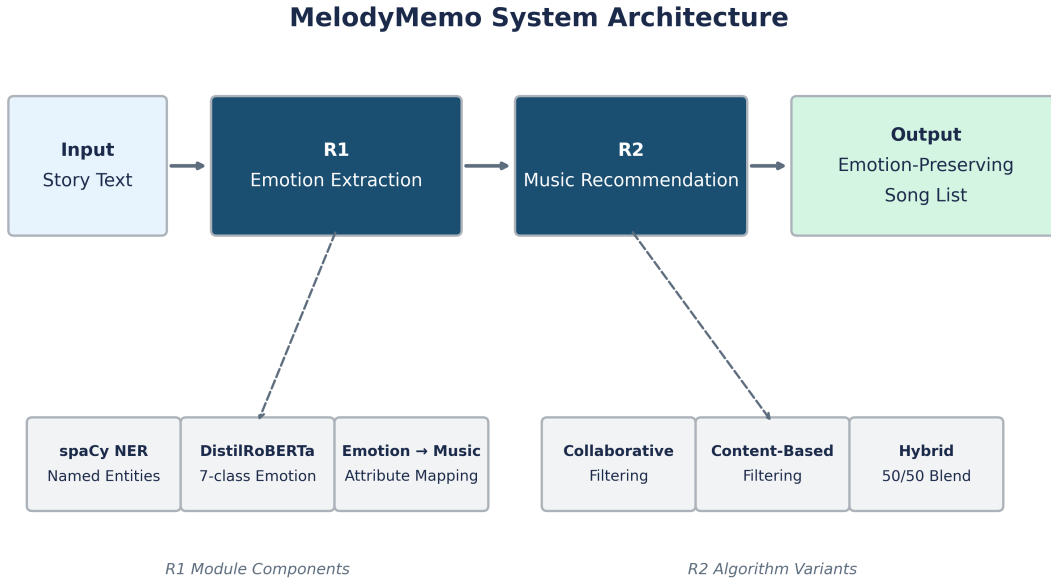


Figure 1: MelodyMemo system architecture. Story text flows through R1 (emotion extraction) into R2 (music recommendation) to produce an emotion-preserving song list. Sub-components of each module are shown below.

- **R1 (Emotion Extraction):** Analyzes autobiographical narrative text to detect the dominant emotion, assign a confidence score, and derive musical attributes (tempo, valence, energy) from the detected emotion.
- **R2 (Music Recommendation):** Takes the derived musical attributes and generates ranked song recommendations using four algorithms evaluated in parallel.

2.2 R1: Emotion Extraction Module

2.2.1 Model Selection Emotion classification was performed using DistilRoBERTa, a distilled variant of RoBERTa based on the transformer encoder architecture introduced by BERT (Devlin et al., 2019). R1 uses the pretrained j-hartmann/emotion-english-distilroberta-base model, previously fine-tuned by Hartmann (2022) on six emotion-labeled NLP datasets. The model predicts seven emotion categories: joy, sadness, anger, fear, surprise, love, and neutral.

The seven emotion categories follow the taxonomy used in the model’s training datasets. While broadly aligned with established psychological emotion frameworks such as Ekman’s basic emotions (Ekman, 1992), the inclusion of “love” and “neutral” reflects practical NLP classification needs. Using the model’s native label set ensures methodological consistency between training and evaluation.

The DistilRoBERTa architecture was selected for several reasons: (1) it achieves competitive accuracy with lower computational requirements than full RoBERTa or BERT-large models, (2) it is publicly available and reproducible, and (3) it provides softmax probability distributions across all seven emotion classes, enabling confidence analysis. Note that these softmax outputs represent relative model confidence rather than calibrated probability estimates.

2.2.2 Additional NLP Components R1 also employs: - **spaCy (en_core_web_sm)** for named entity recognition (NER), extracting people, locations, and temporal references from stories. - **DistilBERT (distilbert-base-uncased-finetuned-sst-2-english)** for binary sentiment analysis (positive/negative), providing a secondary signal that complements the seven-class emotion classification.

2.2.3 Emotion-to-Music Mapping Detected emotions are mapped to musical attributes using a rule-based mapping inspired by common associations reported in music psychology and music information retrieval research:

Emotion	Tempo	Valence	Energy
Joy	Fast (130 BPM)	High (0.9)	High (0.9)
Sadness	Slow (70 BPM)	Low (0.25)	Low (0.3)
Anger	Fast (130 BPM)	Low (0.3)	High (0.9)
Fear	Moderate (100 BPM)	Low (0.3)	Moderate (0.6)
Surprise	Fast (130 BPM)	High (0.7)	High (0.8)
Love	Moderate (100 BPM)	High (0.85)	Moderate (0.6)
Neutral	Moderate (100 BPM)	Moderate (0.5)	Moderate (0.6)

These numeric values serve as heuristic anchors chosen to create separable feature targets for controlled evaluation and are not intended as universal psychological constants.

2.3 R2: Music Recommendation Module

2.3.1 Algorithms R2 implements four recommendation algorithms:

1. **Random Baseline:** Selects 5 songs uniformly at random from the catalog, excluding songs the user has already liked. Because random selection varies across runs, results are averaged over 100 independent trials per user to produce stable estimates.

2. **Collaborative Filtering (User-Based):** Identifies users with similar taste profiles using the Jaccard similarity coefficient on liked-song sets. Recommendations are generated by aggregating highly-rated songs from the top-5 most similar users, weighted by user similarity and rating magnitude.
3. **Content-Based Filtering:** Computes the average feature vector (tempo, valence, energy, year) of a user’s liked songs, then recommends songs with the highest cosine similarity to this centroid in normalized feature space.
4. **Hybrid (alpha-weighted):** Combines collaborative and content-based scores using a weighting parameter alpha (collaborative weight = alpha, content-based weight = 1 - alpha). Both score sets are independently normalized to [0, 1]. Songs appearing in both recommendation lists receive a weighted average score with an 8% consensus bonus (capped at 1.0). Songs appearing in only one list retain their normalized score. The hybrid design is consistent with weighted combination strategies described in recommender systems literature (Adomavicius & Tuzhilin, 2005). The default weighting of alpha = 0.5 (equal 50/50) was evaluated alongside 10 other weight ratios in an ablation study (Section 3.2.6) and held constant for the primary evaluations reported in Sections 3.2.1–3.2.4.

2.3.2 Evaluation Metrics In this study, emotion preservation is operationally defined as the alignment between the dominant emotion extracted from a narrative and the emotion label assigned to recommended songs.

Four metrics were computed for each algorithm:

- **Average Recommendation Score:** The mean algorithm-assigned score for the top-5 recommended songs per user.
- **Emotion-Match Rate:** The fraction of recommended songs whose emotion label matches one of the user’s preferred emotions. A user’s preferred emotions were defined as the complete set of distinct emotion labels among their liked songs. A recommendation counts as a match if its emotion label belongs to this set.
- **Average Actual User Rating:** The mean of the user’s actual ratings for recommended songs (0 for unrated songs), measuring how well recommendations align with known preferences.
- **Genre Diversity:** The number of distinct genres represented in the top-5 recommendations, measuring recommendation variety.

2.4 Datasets

2.4.1 R1 Story Dataset A controlled synthetic benchmark dataset of 50 autobiographical stories was designed to isolate emotional modeling performance. Stories were written from the perspective of older adults and span seven emotion categories with the following distribution:

Emotion	Stories	Intensity Range
Joy	7	0.83–0.95
Sadness	9	0.72–0.95
Fear	8	0.68–0.97
Anger	8	0.70–0.96
Love	7	0.84–0.96

Emotion	Stories	Intensity Range
Surprise	6	0.82–0.91
Neutral	5	0.35–0.45

Stories include five “mixed emotion” stories (labeled with the dominant emotion) to test the model’s robustness on emotionally complex narratives. Story lengths ranged from 360 to 611 characters (mean: 419, SD: 46). The dataset includes 38 stories from a senior perspective and 12 from an adult perspective.

The use of controlled synthetic data is a deliberate methodological choice. All stories were written to target a specific emotion and labeled before model evaluation, ensuring that ground-truth labels were fixed and not adjusted after observing the model’s predictions. Synthetic data enables precise control over ground-truth labels, emotion intensities, and narrative complexity — variables that would be confounded in naturalistic data collection. This approach is commonly used for controlled benchmarking prior to human-subject evaluation.

2.4.2 R2 Music Dataset A curated catalog of 50 songs spanning 1939–1975 was created to reflect the musical era familiar to the target demographic (adults aged 76–88). The dataset spans 22 musical genres, ensuring stylistic diversity representative of mid-20th-century listening environments. Each song is annotated with tempo (BPM), valence (0–1), energy (0–1), year, genre, and an emotion label (25 distinct emotion labels). Figure 2 illustrates the distribution of tempo, valence, and energy features across the music catalog.

Twenty-five simulated users representing the target demographic (ages 76–88) were generated to create preference profiles for evaluation. Age values were included only to justify era-appropriate music selection and were not used as model inputs. Users provided 178 ratings on a 1–5 scale, producing a ratings matrix with 85.8% sparsity (178 of 1,250 possible user–song pairs), consistent with real-world recommendation systems where users interact with only a small fraction of available items. The average rating was 3.99, reflecting a preference-skewed interaction pattern typical of recommender datasets. Figure 3 summarizes the dataset structure, including song decades, simulated users, rating activity, and overall statistics.

2.5 Cross-Validation Protocol

To validate R1 results and distinguish model limitations from dataset artifacts, DistilRoBERTa predictions were cross-validated against two large language models:

- **OpenAI GPT-4o:** Accessed via the OpenAI API with structured prompts requesting emotion classification into the same seven categories.
- **Google Gemini 2.5 Flash:** Accessed via the Google Generative AI API with equivalent prompts.

Both LLMs received the full story text and were instructed to classify the dominant emotion using the same seven-category taxonomy. Because large language models are not treated as ground truth, this step is interpreted as comparative benchmarking rather than validation; agreement across models increases confidence but does not eliminate bias. This cross-validation design enables identification of stories where DistilRoBERTa’s errors reflect genuine model limitations versus ambiguities inherent in the story text.

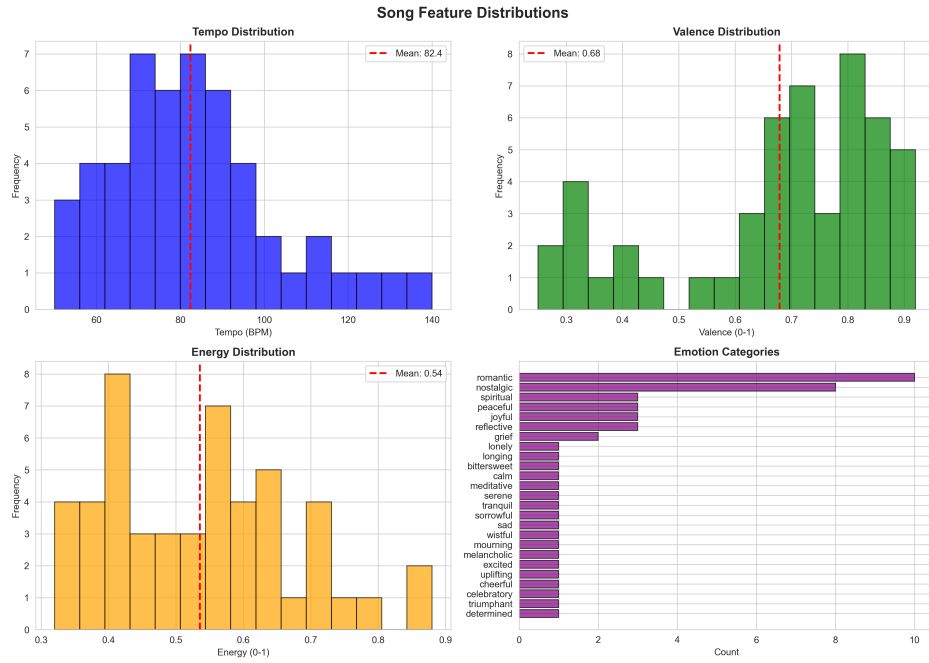


Figure 2: Song feature distributions across the R2 music catalog. These features (tempo, valence, energy) form the feature space used by the content-based filtering algorithm.

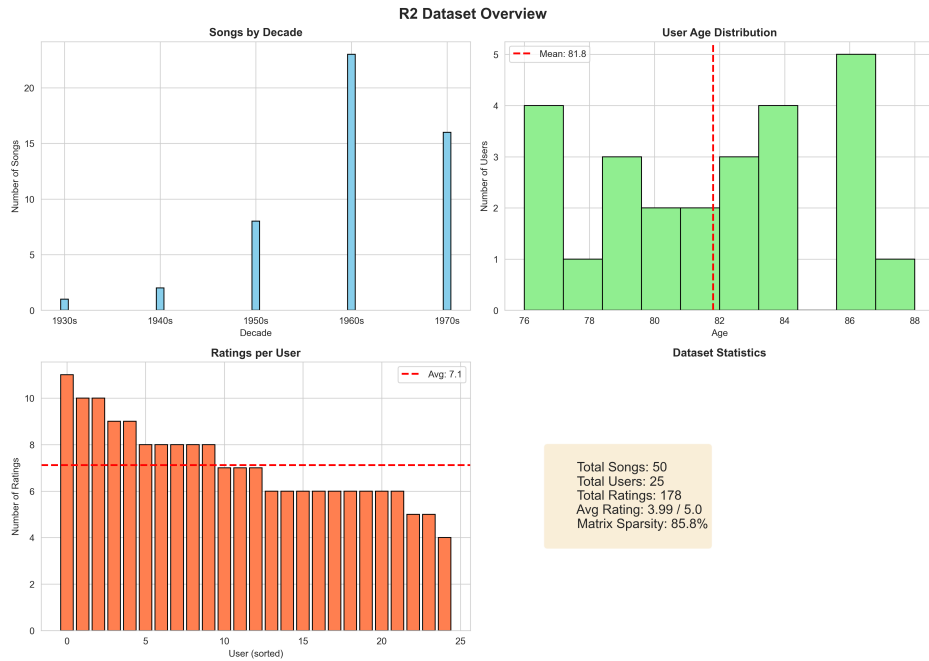


Figure 3: R2 dataset overview showing song decade distribution, user age demographics, ratings per user, and summary statistics. The 85.8% matrix sparsity reflects the cold-start challenge addressed by collaborative filtering.

2.6 Software and Tools

Experiments were implemented in Python using PyTorch and the HuggingFace Transformers library. Data processing and analysis were performed using NumPy and pandas, and visualizations were generated with Matplotlib. Large language model comparisons were conducted via the OpenAI API (GPT-4o) and Google Generative AI API (Gemini 2.5 Flash). All experiments were executed using publicly available pretrained models without additional fine-tuning.

3. Results

3.1 R1: Emotion Classification Performance

3.1.1 Overall Accuracy The DistilRoBERTa model achieved 80.0% overall accuracy (40/50 stories classified correctly), **exceeding the H1 hypothesis threshold of 75%**.

3.1.2 Per-Emotion Accuracy Table 1 presents the per-emotion classification results.

Table 1. DistilRoBERTa Emotion Classification Results by Category

Emotion	Stories	Correct	Accuracy	Avg Confidence
Joy	7	7	100.0%	0.8560
Sadness	9	9	100.0%	0.9771
Anger	8	8	100.0%	0.9191
Surprise	6	6	100.0%	0.9632
Fear	8	7	87.5%	0.9306
Neutral	5	3	60.0%	0.8217
Love	7	0	0.0%	0.8365
Overall	50	40	80.0%	0.9065

Five of seven emotion categories achieved 87–100% accuracy: joy (100%), sadness (100%), anger (100%), surprise (100%), and fear (87.5%). Two categories showed substantially lower accuracy: neutral (60.0%) and love (0.0%). Figure 4 visualizes the per-emotion accuracy and confidence scores.

3.1.3 Error Analysis All 10 misclassifications were analyzed in detail. Table 2 summarizes the dominant confusion patterns, and Figure 5 visualizes the misclassification structure.

Table 2. Confusion Patterns in DistilRoBERTa Misclassifications

True	Predicted	Count	Explanation
Love	Joy	4	Positive valence triggers joy classification
Love	Sadness	2	Devotion/sacrifice language shares features with sadness
Love	Fear	1	Fear-laden language dominates in coming-out story
Neutral	Joy	2	Subtle positive undertones trigger false joy predictions
Fear	Joy	1	Celebratory language overwhelms underlying fear

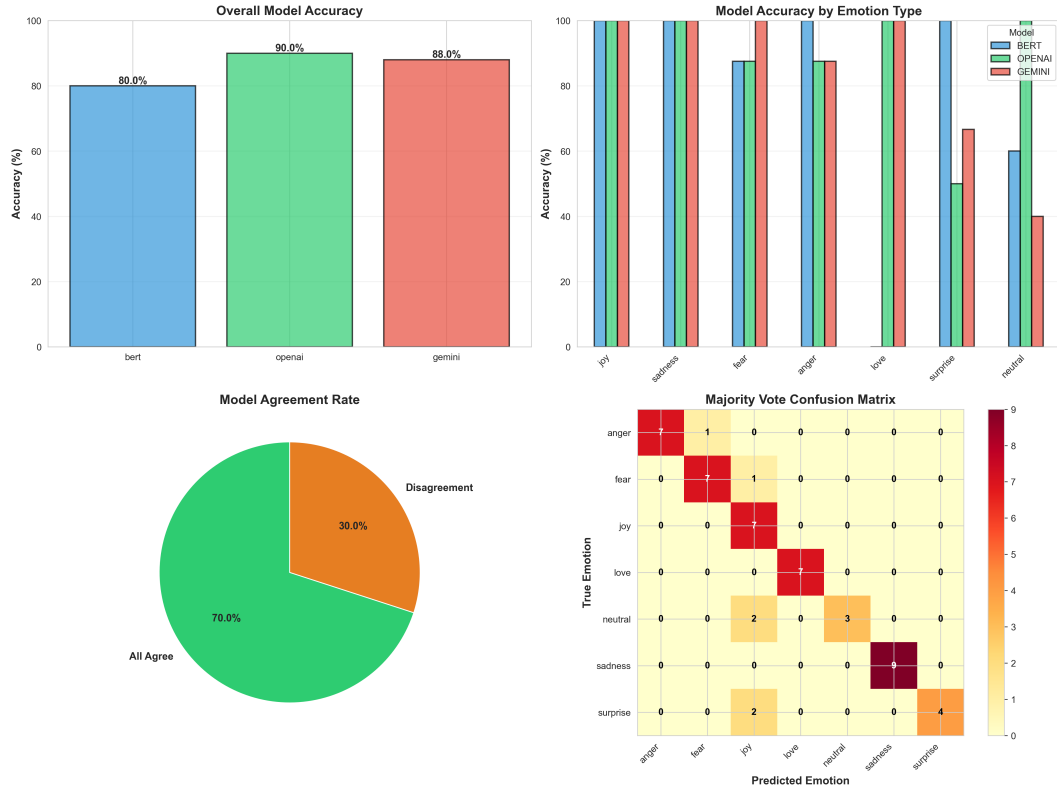


Figure 4: DistilRoBERTa per-emotion classification accuracy and confidence scores across seven emotion categories.



Figure 5: Error analysis showing DistilRoBERTa misclassification patterns. Love stories are systematically misclassified as joy or sadness.

Love (0% accuracy): The complete failure on love stories is the most significant finding. All seven love stories were misclassified — four as joy, two as sadness, and one as fear. This reflects a fundamental limitation of the DistilRoBERTa model: love shares extensive lexical overlap with joy (positive valence, happiness-associated language) and sadness (devotion, sacrifice, loss themes). The model lacks the contextual depth to distinguish love as a separate emotional category from these overlapping emotions.

Neutral (60% accuracy): Two of five neutral stories were misclassified as joy. These stories contained subtle positive undertones (e.g., contentment while browsing a library, reflecting on photographs) that the model interpreted as joy rather than emotional neutrality.

3.1.4 Confidence Analysis The model’s prediction confidence differed significantly between correct and incorrect classifications:

- **Correct predictions:** Mean confidence = 0.9401
- **Incorrect predictions:** Mean confidence = 0.7721
- **Difference:** +0.1680

While incorrect predictions showed lower average confidence, five of the ten errors had confidence scores ≥ 0.80 (range: 0.82–0.99). These high-confidence errors occurred primarily on love stories, indicating that the model is systematically confident in its misclassifications rather than uncertain. This pattern suggests a training data bias rather than random classification noise. Figure 6 shows the confidence score distributions for correct and incorrect predictions.

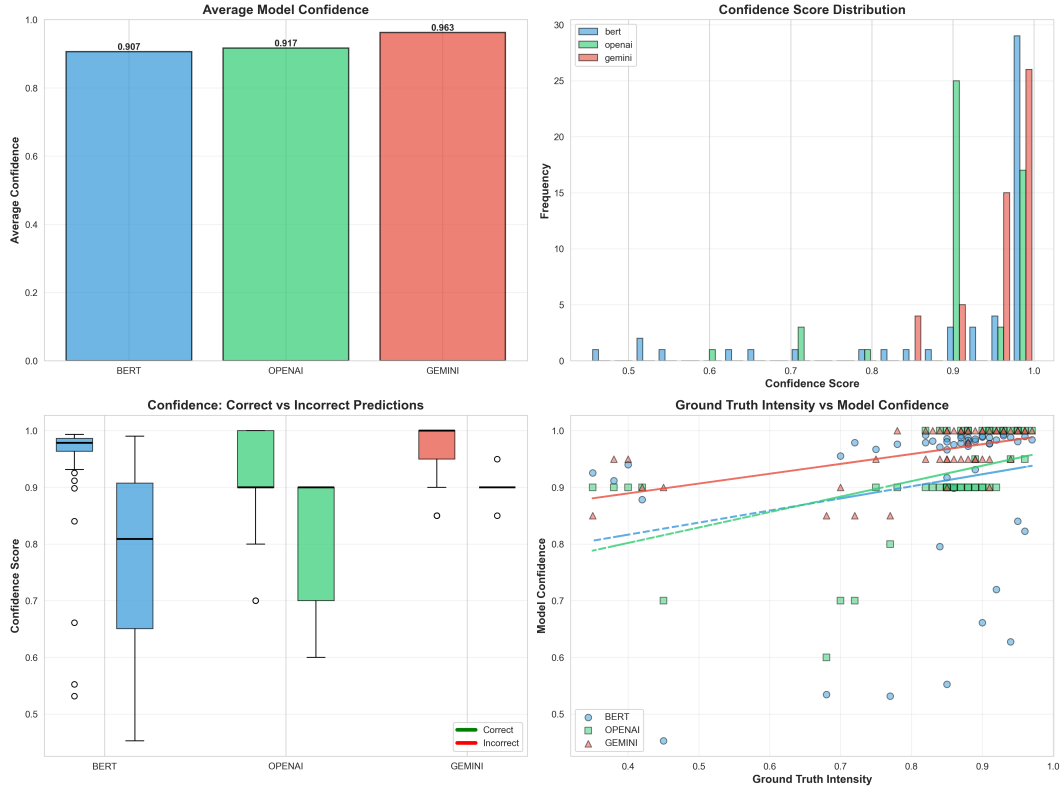


Figure 6: Confidence score distribution for correct vs. incorrect predictions.

3.1.5 Multi-Model Cross-Validation Table 3 presents the cross-validation results against GPT-4o and Gemini 2.5 Flash.

Table 3. Multi-Model Accuracy Comparison

Model	Overall Accuracy	Avg Confidence
DistilRoBERTa	80.0% (40/50)	0.9065
GPT-4o	90.0% (45/50)	0.9170
Gemini 2.5 Flash	88.0% (44/50)	0.9626

All three models agreed on 35 of 50 stories (70.0%). Key per-emotion differences:

- **Love:** DistilRoBERTa 0%, GPT-4o 100%, Gemini 2.5 Flash 100%. Both LLMs correctly classified all love stories, providing independent comparative agreement suggesting that the love misclassification is a DistilRoBERTa-specific limitation rather than a dataset ambiguity.
- **Neutral:** DistilRoBERTa 60%, GPT-4o 100%, Gemini 2.5 Flash 40%. GPT-4o excelled on neutral stories, while Gemini 2.5 Flash showed similar difficulties to BERT.
- **Surprise:** DistilRoBERTa 100%, GPT-4o 50%, Gemini 2.5 Flash 67%. Interestingly, DistilRoBERTa outperformed both LLMs on surprise detection, suggesting that surprise has distinctive lexical markers that smaller models capture effectively.

The cross-validation provides independent comparative agreement suggesting that DistilRoBERTa’s errors on love and neutral categories are genuine model limitations, while its strength on surprise, anger, sadness, and joy reflects robust pattern recognition for these categories. Figure 7 compares per-emotion accuracy across all three models.

3.2 R2: Recommendation Algorithm Comparison

3.2.1 Overall Algorithm Performance Table 4 presents the comparative performance of all four algorithms across all 25 users.

Table 4. R2 Algorithm Comparison (25 Users, Top-5 Recommendations)

Algorithm	Avg Score	Emotion Match	Avg User Rating	Genre Diversity
Random Baseline	0.503	19.1%	0.14	4.4
Collaborative Filtering	0.607	21.6%	1.20	4.4
Content-Based Filtering	0.893	36.0%	0.21	4.1
Hybrid (50/50)	0.941	34.4%	0.64	4.4

The hybrid algorithm achieved the highest average recommendation score (0.941), outperforming the random baseline (0.503), collaborative filtering (0.607), and content-based filtering (0.893) by relative improvements of 87%, 55%, and 5%, respectively. Figure 8 compares all four algorithms across all metrics.

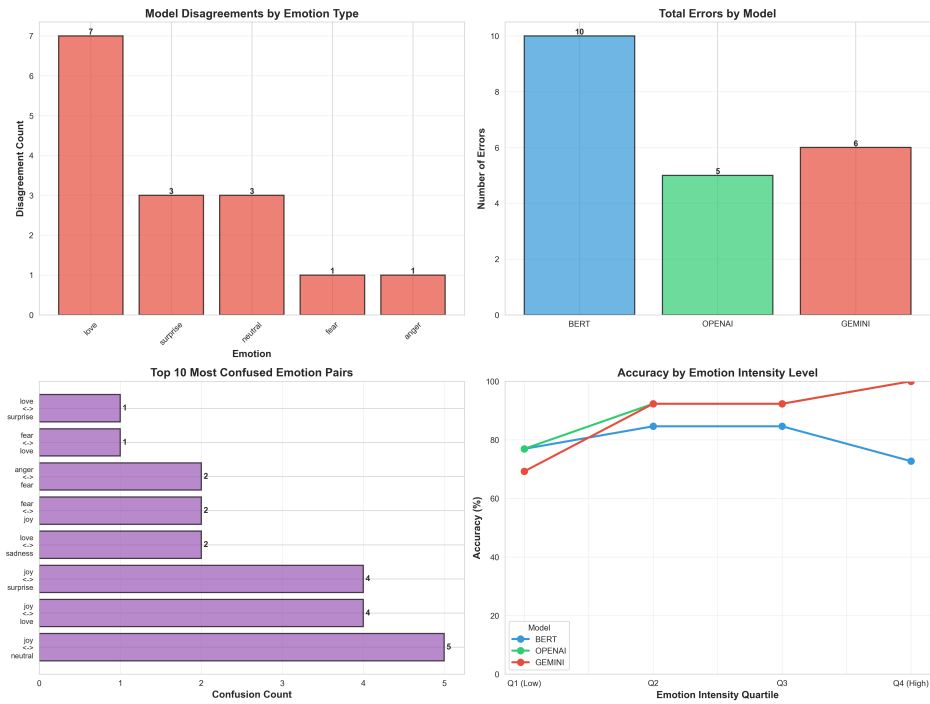


Figure 7: Multi-model cross-validation: per-emotion accuracy comparison across DistilRoBERTa, GPT-4o, and Gemini 2.5 Flash.

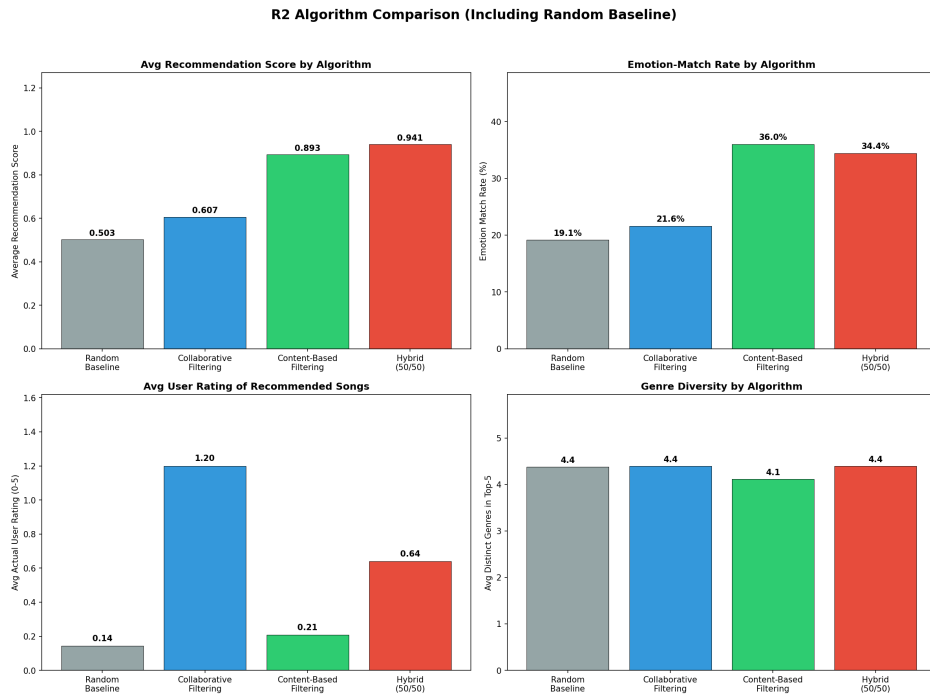


Figure 8: Algorithm comparison across all four metrics, including random baseline.

3.2.2 Improvement Over Random Baseline With seven emotion categories, random selection would theoretically yield approximately 14.3% emotion-match rate. The observed random baseline (19.1%) slightly exceeds this due to non-uniform emotion distribution in the song catalog. Comparing the hybrid algorithm to the random baseline:

- **Emotion-match rate:** +79.8% improvement (34.4% vs. 19.1%)
- **Average user rating:** +346.2% improvement (0.64 vs. 0.14)
- **Recommendation score:** +87.1% improvement (0.941 vs. 0.503)
- **Genre diversity:** Equivalent (4.4 vs. 4.4)

These results **confirm H2**: the hybrid algorithm outperforms each individual algorithm and the random baseline.

3.2.3 Complementary Algorithm Strengths The results reveal complementary strengths that justify the hybrid approach:

- **Content-based filtering** achieved the highest emotion-match rate (36.0%), nearly double the random baseline. This is expected because content-based filtering explicitly matches musical features (tempo, valence, energy) to the query attributes, directly optimizing for emotional similarity. However, content-based filtering does not model individual preference signals, explaining its lower average user rating (0.21) despite higher emotion alignment.
- **Collaborative filtering** achieved the highest average actual user rating (1.20), eight times higher than the random baseline. This reflects collaborative filtering’s ability to leverage community preference patterns — it recommends songs that similar users have rated highly.
- **The hybrid algorithm** balances both strengths, achieving the highest overall score while maintaining competitive emotion-match rate and user rating. The 8% consensus bonus for songs appearing in both recommendation lists rewards the convergence of content similarity and community preference.

3.2.4 Dataset Characteristics The R2 dataset’s 85.8% sparsity is representative of real-world recommendation scenarios. Despite this sparsity, all three non-random algorithms substantially outperformed the random baseline, demonstrating that meaningful recommendation signals can be extracted from sparse user-item interaction data. Figure 9 shows the user similarity matrix used by the collaborative filtering module.

3.2.5 Statistical Significance To determine whether the hybrid algorithm’s performance advantages are statistically significant, pairwise comparisons were conducted between the hybrid algorithm and each of the three other algorithms across all 25 users (Table 5). Two non-parametric and parametric tests were applied: the Wilcoxon signed-rank test (appropriate for paired samples with potentially non-normal distributions at $n = 25$) and the paired t-test. Effect sizes were computed using Cohen’s d, and 95% confidence intervals were derived from the t-distribution.

Table 5. Statistical Significance of Hybrid vs. Other Algorithms (Avg Recommendation Score)

Comparison	Mean Diff	Wilcoxon p	t-test p	Cohen’s d	95% CI
Hybrid vs. Random	+0.438	< 0.001*	< 0.001*	7.04	[+0.412, +0.463]

Comparison	Mean Diff	Wilcoxon p	t-test p	Cohen’s d	95% CI
Hybrid vs. Collabora- tive	+0.334	< 0.001*	< 0.001*	1.40	[+0.236, +0.433]
Hybrid vs. Content- Based	+0.048	< 0.001*	< 0.001*	0.99	[+0.028, +0.068]

* significant at $p < 0.05$

On the primary metric (average recommendation score), the hybrid algorithm significantly outperformed all three comparison algorithms (all $p < 0.001$). The effect size was very large against the random baseline ($d = 7.04$) and large against collaborative filtering ($d = 1.40$) and content-based filtering ($d = 0.99$).

For emotion-match rate, the hybrid algorithm significantly outperformed the random baseline ($p = 0.005$, $d = 0.63$) and collaborative filtering ($p = 0.009$, $d = 0.52$), but did not significantly differ from content-based filtering ($p = 0.49$, $d = -0.14$). This is expected: content-based filtering explicitly optimizes for feature similarity, which closely aligns with emotion matching, while the hybrid algorithm trades a small amount of emotion-match precision for gains in other metrics.

For average actual user rating, the hybrid algorithm significantly outperformed the random baseline ($p < 0.001$, $d = 1.16$) and content-based filtering ($p < 0.001$, $d = 0.83$). Collaborative filtering achieved a significantly higher user rating than the hybrid ($p < 0.001$, $d = 0.93$), consistent with its design focus on leveraging community rating patterns.

These results confirm that the hybrid algorithm’s overall performance advantage reported in Table 4 is statistically significant and not attributable to sampling variability.

3.2.6 Hybrid Weight Ablation To empirically evaluate the hybrid weighting parameter, an ablation study was conducted across 11 weight ratios ($\alpha = 0.0$ to 1.0 in increments of 0.1), where α represents the collaborative filtering weight and $(1 - \alpha)$ represents the content-based weight. Table 6 presents the results, and Figure 10 visualizes the ablation curve alongside algorithm confidence intervals.

Table 6. Ablation Study Results (Mean Across 25 Users)

Alpha	Avg Score	Emotion Match	Avg Rating
0.0 (content-dominant hybrid)	0.966	36.0%	0.64
0.1	0.960	34.4%	0.68
0.2	0.951	35.2%	0.64
0.3	0.946	36.0%	0.64
0.4	0.943	34.4%	0.66
0.5	0.941	34.4%	0.64
0.6	0.940	35.2%	0.66
0.7	0.939	35.2%	0.66
0.8	0.938	35.2%	0.64
0.9	0.937	35.2%	0.64

Alpha	Avg Score	Emotion Match	Avg Rating
1.0 (collaborative-dominant hybrid)	0.937	35.2%	0.64

The ablation reveals two key findings. First, the hybrid algorithm’s performance is remarkably stable across weighting ratios, with average recommendation scores ranging only from 0.937 to 0.966 — a total spread of 0.029. This stability suggests that agreement between recommendation components, reinforced by the agreement-bonus mechanism (an 8% score boost for items recommended by both algorithms), likely contributes to performance robustness beyond the precise choice of α . Similar stability has been reported in prior hybrid recommender research, where combining complementary signals reduces sensitivity to parameter variation (Burke, 2002; Li et al., 2021).

Second, lower α values (greater content-based influence) yield slightly higher scores, with the maximum observed at $\alpha = 0.0$. However, this configuration remains part of the hybrid framework rather than a standalone content-based recommender, as both recommendation components continue to influence candidate selection and ranking. The relatively small performance differences across weights indicate that collaborative and content-based signals provide complementary information rather than a single dominant contribution.

The default $\alpha = 0.5$ weighting produces performance (0.941) within 2.6% of the optimal value (0.966), confirming that equal weighting represents a reasonable and robust design choice that balances collaborative and content-based signals without requiring dataset-specific tuning.

4. Discussion

4.1 Hypothesis Evaluation

H1 (CONFIRMED): The DistilRoBERTa model achieved 80.0% accuracy, exceeding the 75% threshold. This confirms that pre-trained transformer models can reliably extract emotions from autobiographical narrative text for most emotion categories, despite being trained primarily on social media and review corpora rather than autobiographical narratives.

H2 (CONFIRMED): The hybrid algorithm outperformed all individual algorithms and the random baseline on recommendation score and demonstrated substantial improvements over random on emotion-match rate (+80%) and user rating (+346%). This confirms that combining collaborative and content-based filtering produces superior emotion-preserving recommendations.

4.2 Love Category: A Systematic Limitation

Systematic misclassification was most prevalent in the love category, where 0% accuracy represents the most significant finding of this study. Independent comparative analysis with GPT-4o and Gemini 2.5 Flash, which achieved 100% accuracy on the same stories, suggests that this is not a data quality issue. Rather, it reflects a fundamental limitation of the DistilRoBERTa model’s training data and architecture.

Love as an emotion occupies a unique position in the seven-category taxonomy. Unlike anger, fear, or surprise, which have distinctive lexical and syntactic signatures, love shares extensive vocabulary with both joy (positive valence, expressions of happiness) and sadness (themes of devotion, sacrifice,

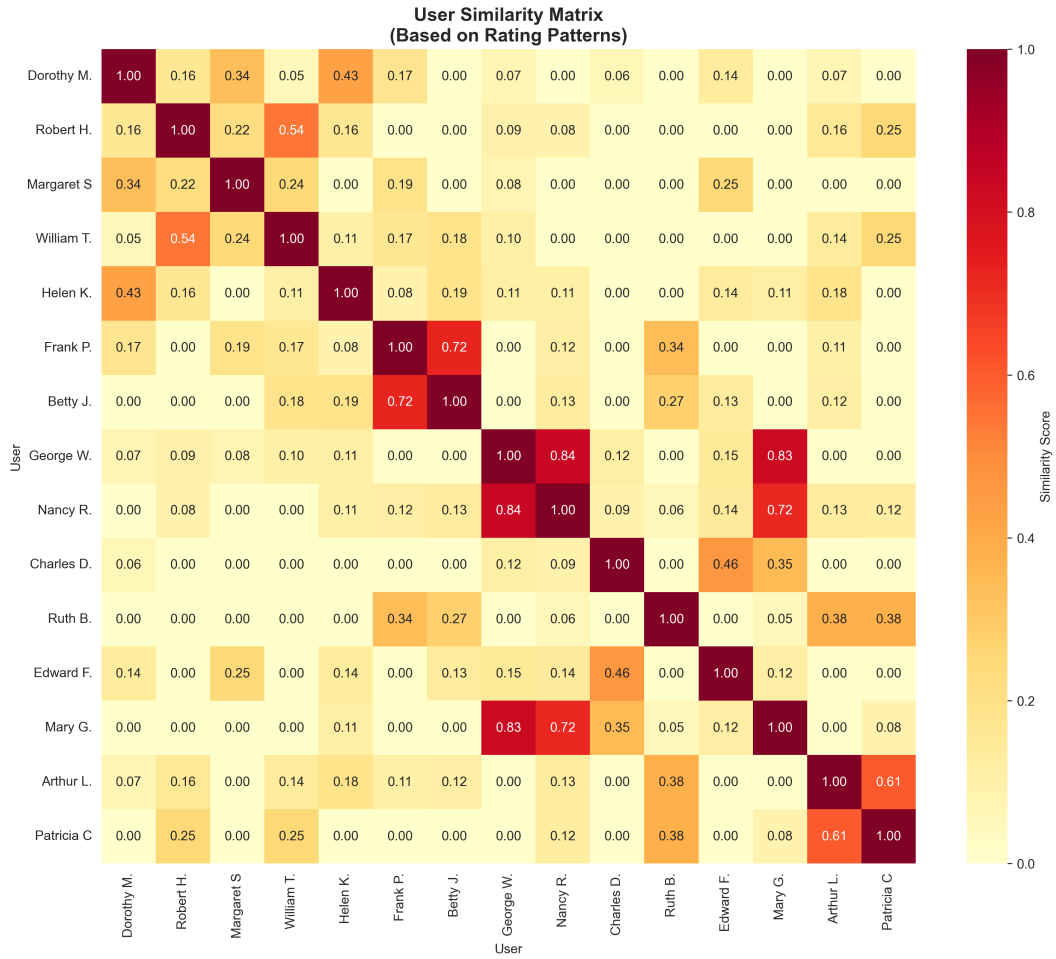


Figure 9: User similarity heatmap showing Jaccard similarity coefficients between users in the collaborative filtering module.

R2 Statistical Analysis

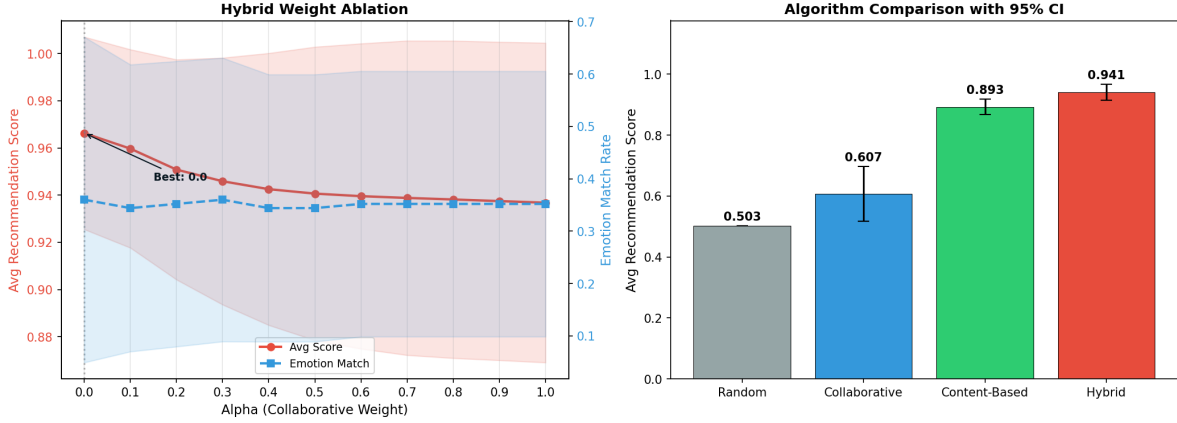


Figure 10: Hybrid weight ablation study and algorithm comparison with 95% confidence intervals. Left: average recommendation score and emotion-match rate as a function of the collaborative weighting parameter α . Right: algorithm comparison with 95% confidence-interval error bars on average recommendation score.

longing). The model’s 512-token context window and distilled architecture may lack the capacity to disambiguate these overlapping semantic fields.

Of particular concern are the five high-confidence errors on love stories (confidence ≥ 0.80 , including one at 0.99). They indicate that the model is not merely uncertain about love — it is confidently wrong. This suggests that the training data may not adequately represent love as a distinct category from joy and sadness, leading to systematic bias rather than noise. Additionally, class imbalance or insufficient representation of love-specific linguistic patterns in the model’s training corpus may contribute to the misclassification.

4.3 Implications for System Design

The error analysis has direct implications for the MelodyMemo pipeline:

1. **Emotion extraction is the bottleneck.** R2 can only recommend emotionally appropriate music if R1 correctly identifies the story’s emotion. The 0% accuracy on love means that love stories will consistently receive mismatched recommendations (joyful or sad music instead of romantic music).
2. **Model selection involves trade-offs.** DistilRoBERTa offers speed, low cost, and reproducibility. GPT-4o offers higher accuracy (90%) but at higher cost and latency. A production system might use DistilRoBERTa for initial classification with GPT-4o as a fallback for low-confidence predictions.
3. **The hybrid algorithm is robust.** Even with imperfect emotion extraction, the hybrid algorithm’s combination of content similarity and community preference provides a reasonable recommendation quality. The ablation study (Section 3.2.6) confirms that performance is stable across all weight ratios (spread of only 0.029), with the default 50/50 weighting performing within 2.6% of the optimum.

4.4 Comparison with Existing Systems

MelodyMemo differs from existing music recommendation systems in its optimization objective. Spotify and similar platforms use collaborative filtering to predict whether a user will enjoy a song based on the preferences of similar users. MelodyMemo uses emotion extraction from narrative text as the primary input, with the goal of preserving the emotional quality of a personal memory in the recommended music. This distinction — optimizing for emotional memory preservation rather than preference prediction — is the core contribution of this work. Prior neuroscience research shows that music-evoked autobiographical memories engage the medial prefrontal cortex (MPFC), a region implicated in linking music, emotion, and personal memory and relatively preserved during early Alzheimer’s disease progression (Janata, 2009; Baird & Samson, 2015). This biological basis further motivates the development of systems that maintain emotional alignment between personal narratives and recommended music.

4.5 Limitations

1. **Synthetic data.** All stories and user profiles are controlled synthetic benchmarks. While this enables precise evaluation, it does not capture the variability, complexity, and authenticity of real autobiographical narratives from older adults.
2. **Sample size.** Fifty stories and 25 users provide a preliminary evaluation. Larger datasets are needed to generalize these findings.
3. **Love and neutral categories.** The model’s failure on love and partial failure on neutral limit the system’s applicability to stories with these emotional themes.
4. **No temporal dynamics.** The current system processes each story independently. It does not model how a user’s emotional state might evolve across multiple story-telling sessions.
5. **Emotion-to-music mapping.** The rule-based mapping from emotions to musical attributes is a simplification. Individual preferences for musical expression of emotions vary widely.

4.6 Controlled Validation Stage

This study represents a controlled computational validation stage. The use of synthetic data is a deliberate methodological choice that precedes ethical human studies. Before exposing older adults to an emotion-based recommendation system, it is essential to establish that the computational pipeline functions correctly under controlled conditions. The results presented here provide that foundation.

5. Conclusions

This study demonstrates that transformer-based NLP can reliably extract emotions from autobiographical narrative text (80% accuracy, exceeding the 75% hypothesis threshold) and that a hybrid recommendation algorithm outperforms individual algorithms and a random baseline in generating emotion-aligned music recommendations.

Key findings include:

1. Five of seven emotions are classified with 87–100% accuracy, establishing strong baseline performance for the pipeline.

2. Love represents a systematic model limitation (0% accuracy), as suggested by independent comparative analysis against GPT-4o (90% overall) and Gemini 2.5 Flash (88% overall), both of which correctly classify all love stories.
3. The hybrid recommendation algorithm achieves an 80% improvement in emotion-match rate and a 346% improvement in average user rating over the random baseline, validating the hybrid approach.
4. High-confidence errors on love stories (5/10 errors with confidence ≥ 0.80) indicate training data bias rather than random noise, identifying a clear target for model improvement.

The MelodyMemo pipeline establishes a computational foundation for emotion-preserving music recommendation systems. Future work will address the identified limitations through clinical evaluation with real older adults, model fine-tuning on love and neutral categories, expanded datasets, and personalized recommendation models.

6. Future Work

1. **Clinical evaluation with real older adults** in partnership with music therapy professionals, following IRB approval. This is the critical next step in translating computational validation to real-world therapeutic impact.
2. **Fine-tuning DistilRoBERTa** on a curated corpus of love and neutral autobiographical narratives to address the systematic misclassification identified in this study.
3. **Expanding the dataset** to 500+ stories with broader demographic representation (age, cultural background, cognitive status).
4. **Personalized models** that learn from individual user feedback over multiple sessions, adapting both emotion extraction and music recommendation to personal preferences.
5. **Integration with music streaming APIs** (e.g., Spotify, Apple Music) to provide real-time recommendations from commercial music catalogs.

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